HACKERS DO BEM

FUNDAMENTAL - TURMA 20

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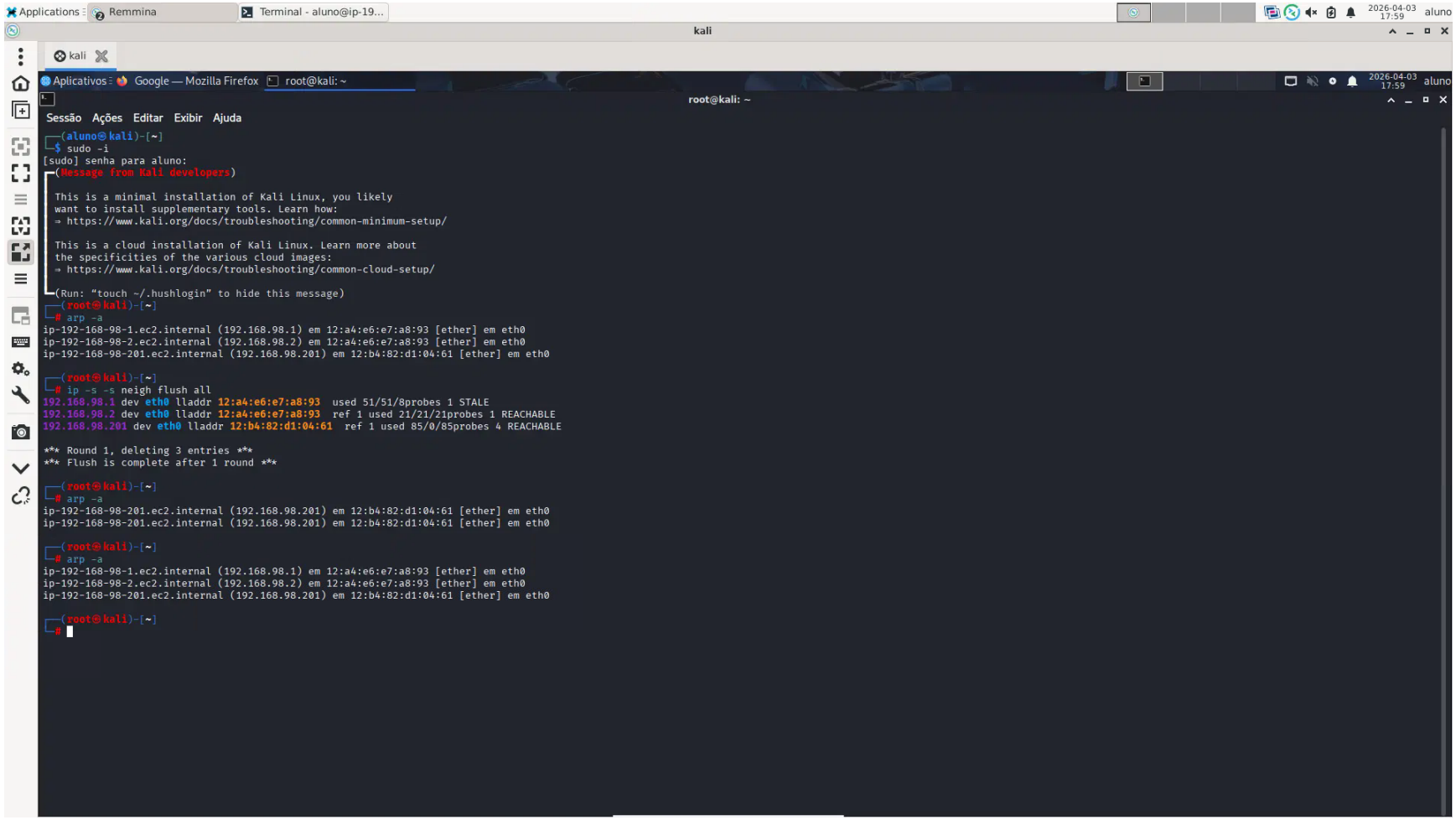
MÓDULO 11

ATIVIDADE PRÁTICA 1

Atividade 11.1: passo 6

Exibição da tabela ARP após acessar o site Google e executar o comando arp -a.



Atividade 11.2: passo 8

Conexão SSH realizada com sucesso para o alias da interface eth0 utilizando o IP 192.168.98.42.

```
64 bytes from 192.168.98.41: icmp_seq=17 ttl=64 time=0.046 ms
64 bytes from 192.168.98.41: icmp_seq=18 ttl=64 time=0.043 ms
^C
--- 192.168.98.41 ping statistics ---
18 packets transmitted, 18 received, 0% packet loss, time 17399ms
rtt min/avg/max/mdev = 0.043/0.078/0.571/0.119 ms

┌──(root㉿kali)-[~]
└─# ping 192.168.98.42
PING 192.168.98.42 (192.168.98.42) 56(84) bytes of data.
64 bytes from 192.168.98.42: icmp_seq=1 ttl=64 time=0.054 ms
64 bytes from 192.168.98.42: icmp_seq=2 ttl=64 time=0.050 ms
64 bytes from 192.168.98.42: icmp_seq=3 ttl=64 time=0.047 ms
64 bytes from 192.168.98.42: icmp_seq=4 ttl=64 time=0.031 ms
64 bytes from 192.168.98.42: icmp_seq=5 ttl=64 time=0.043 ms
64 bytes from 192.168.98.42: icmp_seq=6 ttl=64 time=0.050 ms
^C
--- 192.168.98.42 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5112ms
rtt min/avg/max/mdev = 0.031/0.045/0.054/0.007 ms

┌──(root㉿kali)-[~]
└─# ssh aluno@192.168.98.42
The authenticity of host '192.168.98.42 (192.168.98.42)' can't be established.
ED25519 key fingerprint is: SHA256:VdqMlT8r9e4dNnZzEKJOWMpA0TkR14C35/JrR1/SZsk
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.98.42' (ED25519) to the list of known hosts.
aluno@192.168.98.42's password:
Linux kali 6.16.8+kali-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.16.8-1kali1 (2025-09-24) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Mon Feb  9 22:05:09 2026 from 192.168.98.201
┌──(Message from Kali developers)
│
│  This is a minimal installation of Kali Linux, you likely
│  want to install supplementary tools. Learn how:
│  ⇒ https://www.kali.org/docs/troubleshooting/common-minimum-setup/
│
│  This is a cloud installation of Kali Linux. Learn more about
│  the specificities of the various cloud images:
│  ⇒ https://www.kali.org/docs/troubleshooting/common-cloud-setup/
└──(Run: "touch ~/.hushlogin" to hide this message)
┌──(aluno㉿kali)-[~]
└─$ exit
sair
Connection to 192.168.98.42 closed.

┌──(root㉿kali)-[~]
└─#
```

Atividade 11.3: passo 8

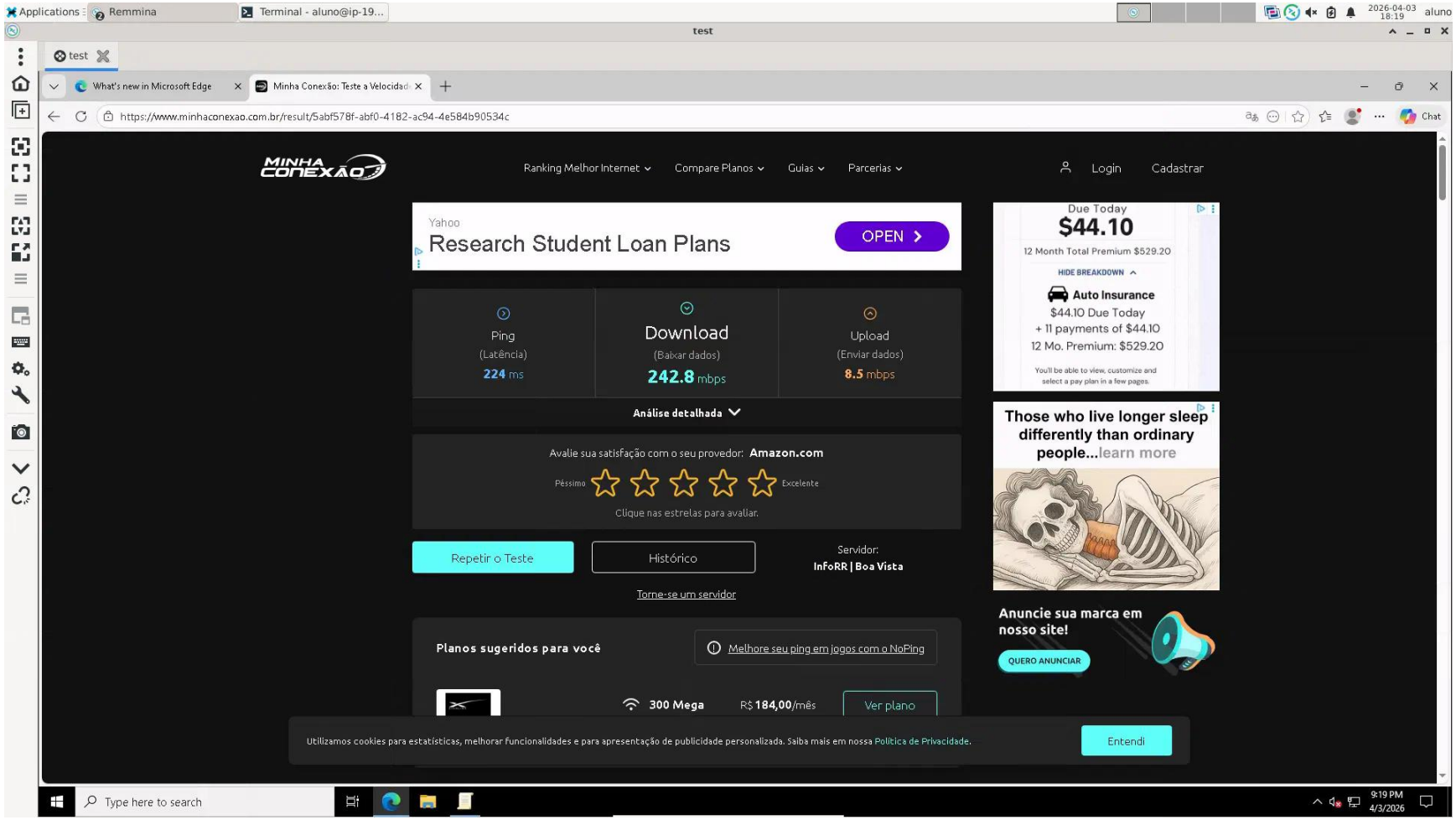
Restauração do endereço MAC original da interface docker0 após o processo de MAC Cloning.

```
Applications  Remmina  Terminal - aluno@ip-19...                    2026-04-03 18:09  aluno
kali
kali
Aplicativos  root@kali: ~                                           2026-04-03 18:09  aluno
root@kali: ~
Sessão  Ações  Editar  Exibir  Ajuda
        inet 192.168.98.40  netmask 255.255.255.0  broadcast 192.168.98.255
        inet6 fe80::1072:72ff:fe5f:19ed  prefixlen 64  scopeid 0x20<link>
        ether 12:72:72:5f:19:ed  txqueuelen 1000  (Ethernet)
        RX packets 2970  bytes 238210 (232.6 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 2552  bytes 2053046 (1.9 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Loopback Local)
        RX packets 25  bytes 1986 (1.9 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 25  bytes 1986 (1.9 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

┌──(root㉿kali)-[~]
└─# ip link set docker0 down

┌──(root㉿kali)-[~]
└─# ip link set dev docker0 address 08:00:27:8b:c0:05

┌──(root㉿kali)-[~]
└─# ip link set docker0 up

┌──(root㉿kali)-[~]
└─# ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP mode DEFAULT group default qlen 1000
    link/ether 12:72:72:5f:19:ed brd ff:ff:ff:ff:ff:ff
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN mode DEFAULT group default
    link/ether 08:00:27:8b:c0:05 brd ff:ff:ff:ff:ff:ff

┌──(root㉿kali)-[~]
└─# ip link set docker0 down

┌──(root㉿kali)-[~]
└─# ip link set dev docker0 address 02:42:40:14:ac:68

┌──(root㉿kali)-[~]
└─# ip link set docker0 up

┌──(root㉿kali)-[~]
└─# ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP mode DEFAULT group default qlen 1000
    link/ether 12:72:72:5f:19:ed brd ff:ff:ff:ff:ff:ff
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN mode DEFAULT group default
    link/ether 02:42:40:14:ac:68 brd ff:ff:ff:ff:ff:ff

┌──(root㉿kali)-[~]
└─#
```

Atividade 11.4: passo 8

Limitação da largura de banda da interface eth0 para 1mbit utilizando controle de tráfego (tc).

```
┌──(root㉿kali)-[~]
└─#

┌──(root㉿kali)-[~]
└─# tc qdisc del dev eth0 root

┌──(root㉿kali)-[~]
└─# ping 192.168.98.1
PING 192.168.98.1 (192.168.98.1) 56(84) bytes of data.
64 bytes from 192.168.98.1: icmp_seq=1 ttl=64 time=0.116 ms
64 bytes from 192.168.98.1: icmp_seq=2 ttl=64 time=0.094 ms
64 bytes from 192.168.98.1: icmp_seq=3 ttl=64 time=0.092 ms
^C
--- 192.168.98.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2031ms
rtt min/avg/max/mdev = 0.092/0.100/0.116/0.010 ms

┌──(root㉿kali)-[~]
└─# tc qdisc add dev eth0 root handle 1: htb

┌──(root㉿kali)-[~]
└─# tc class add dev eth0 parent 1: classid 1:1 htb rate 10mbit

┌──(root㉿kali)-[~]
└─# tc filter add dev eth0 parent 1: protocol ip prio 1 u32 match ip src 192.168.98.1 flowid 1:1

┌──(root㉿kali)-[~]
└─# tc qdisc del dev eth0 root

┌──(root㉿kali)-[~]
└─# tc qdisc add dev eth0 root tbf rate 1mbit burst 10kb latency 50ms

┌──(root㉿kali)-[~]
└─#
```

Atividade 11.5: passo 14

Teste de velocidade após aplicação da política de QoS no Windows Server 2022 com limitação de upload.